

# Inequalities – 6

## Probabilistic and combinatorial inequalities

### Homework

10 сентября 2026 года

**Theoretical minimum.** Markov, Chebyshev, Chernoff and Hoeffding as three applications of one idea; the second moment method; the probabilistic method in its two forms; double counting combined with convexity; Cauchy–Bunyakovsky in counting problems.

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1. (Iranian competition) Let  $X = X_1 + \dots + X_{2m}$ , where the  $X_i$  are independent and take the values 0 and 1 with probability  $\frac{1}{2}$  each. Using Chebyshev's inequality prove that  $\sum_{|k| < \sqrt{m}} \binom{2m}{m+k} \geq 2^{2m-1}$ .
2. (IMC) Let  $n > d$  be positive integers and choose  $n$  independent uniformly distributed points in the unit ball  $B \subset \mathbb{R}^d$ . For  $p \in B$  let  $f(p)$  be the probability that the convex hull of the points contains  $p$ . Prove that  $f(p) \geq f(q)$  whenever  $|p| < |q|$ .
3. (S.T. Yau CSMC 2016) Let  $X_1, X_2, \dots$  be independent identically distributed real random variables. Prove or disprove: if  $\limsup_n \frac{|X_n|}{n} \leq 1$  almost surely, then  $\sum_{n \geq 1} \Pr(|X_n| \geq n) < \infty$ .
4. Prove that a graph on  $n$  vertices with  $m$  edges contains at least  $\frac{4m}{3n} \left(m - \frac{n^2}{4}\right)$  triangles, and identify the two ingredients: double counting and convexity.
5. Prove the Cauchy–Bunyakovsky bound for the number of incidences: if a family of  $m$  subsets of an  $n$ -set has all pairwise intersections of size at most 1, then  $m \leq n^2$ ; and give an example showing the order is right.

6. (Putnam 2023) For  $m \geq 3$  a list of  $\binom{m}{3}$  reals  $a_{ijk}$  is called area definite for  $\mathbb{R}^n$  if  $\sum_{i < j < k} a_{ijk} \text{Area}(\triangle A_i A_j A_k) \geq 0$  for every choice of  $m$  points in  $\mathbb{R}^n$ . Prove that a list which is area definite for  $\mathbb{R}^2$  is area definite for  $\mathbb{R}^3$ .
7. Prove the union bound and the Lovász local lemma in its symmetric form, and use the latter to show that a  $k$ -uniform hypergraph with fewer than  $2^{k-1}$  edges is two-colourable.
8. Prove Chebyshev's inequality for a sum of pairwise independent variables and show that pairwise independence is not enough for Chernoff's bound.
9. Let  $A$  be a set of  $n$  reals. Prove that the number of pairs with difference in a fixed set  $D$  of size  $d$  is at most  $nd$ , and use the second moment method to show that some real number is the difference of at least  $\frac{n(n-1)}{2|A-A|}$  pairs.
10. Prove Jensen's inequality for a random variable and deduce the entropy bound  $H(X) \leq \log |\text{supp } X|$ , with equality exactly for the uniform distribution.